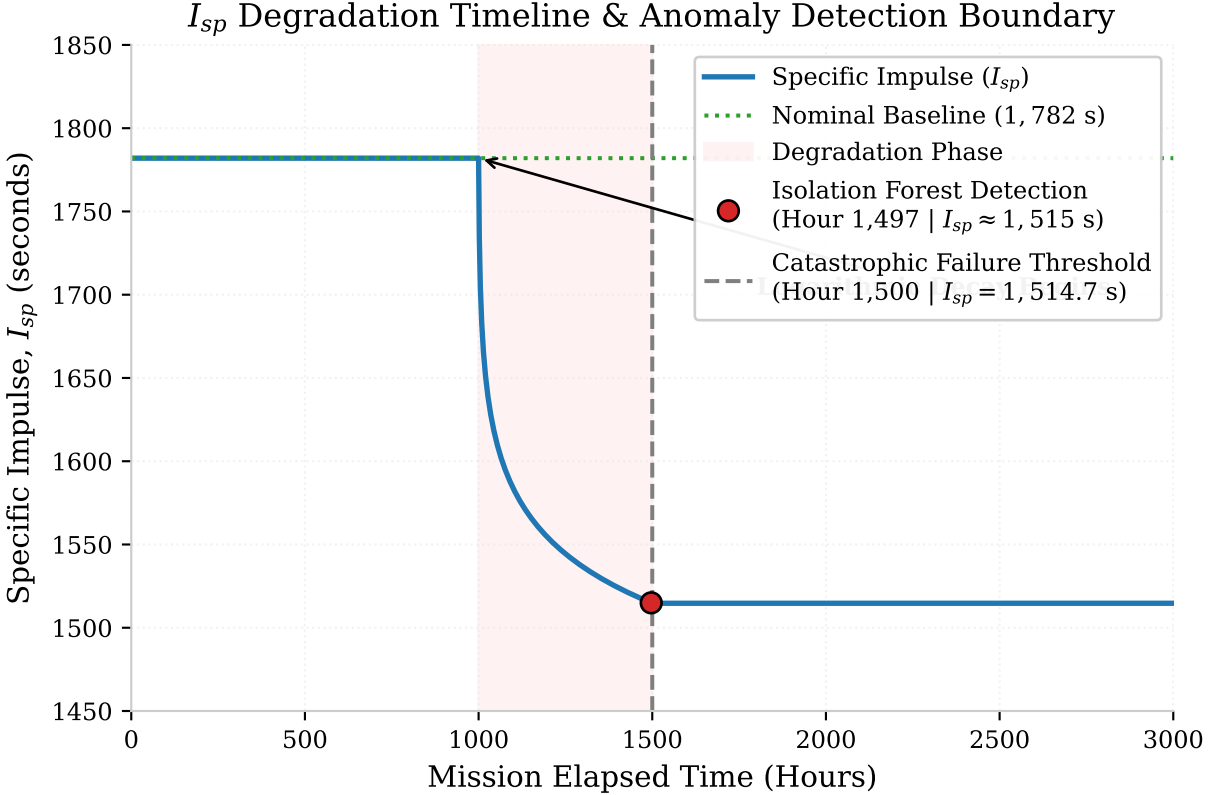
I_{sp} Degradation Timeline & Anomaly Detection Boundary
Specific Impulse (I_{sp})
Nominal Baseline (1, 782 s)
Degradation Phase
Isolation Forest Detection
(Hour 1,497 | $I_{sp} \approx 1,515$ s)
Catastrophic Failure Threshold
(Hour 1,500 | $I_{sp} = 1,514.7$ s)
Specific Impulse, I_{sp} (seconds)
Mission Elapsed Time (Hours)
1850
1800
1750
1700
1650
1600
1550
1500
1450
0
500
1000
1500
2000
2500
3000